Midterm 2 Free Response Solutions (By Version)

Contents

Midterm 2 Free Response Solutions (Translated)

Version A Solutions

Question 22: Bob & Linda

a. **Absolute Advantage (Burgers): Linda** has the absolute advantage because she can produce **40 burgers** per shift, which is more than Bob's **20 burgers**.

b. **Bob's Opportunity Cost of 1 Burger:** Bob gives up 60 cookies to make 20 burgers. Calculation: $\frac{60 \text{ cookies}}{20 \text{ burgers}} = 3$ cookies per burger

c. **Comparative Advantage (Cookies): Bob** has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{3}$ burger (20/60), while Linda's OC is $\frac{1}{2}$ burger (40/80).

d. **Range of Prices (Terms of Trade):** Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 3 cookies). Range: **Between 2 and 3 cookies per burger**.

Question 23: Pollution Permits

a. **Initial Total Cost:** Combined production cost for the first 6 widgets: \$10 + \$25 + \$15 + \$30 + \$20 + \$50 = **\$150**.

b. **WTP and WTA:**

- **Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$60 - \$40 = **\$20**.
- **Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$60 - \$50 = **\$10**.

c. **After-Trade Total Cost:** Cost is minimized by using the 6 permits on the lowest marginal costs available: 10, 15, 20, 25, 30, and 40. New Total Cost: **$140**.

Version B Solutions

Question 22: Bob & Linda

a. **Absolute Advantage (Burgers): Linda** has the absolute advantage because she can produce **30 burgers** per shift, which is more than Bob's **10 burgers**.

b. **Bob's Opportunity Cost of 1 Burger:** Bob gives up 40 cookies to make 10 burgers. Calculation: $\frac{40\text{ cookies}}{10\text{ burgers}}$ = 4 cookies per burger

c. **Comparative Advantage (Cookies): Bob** has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{4}$ burger (10/40), while Linda's OC is $\frac{1}{3}$ burger (30/90).

d. **Range of Prices (Terms of Trade):** Trade will occur if the price of 1 burger is between the two workers' opportunity costs (3 and 4 cookies). Range: **Between 3 and 4 cookies per burger**.

Question 23: Pollution Permits

a. **Initial Total Cost:** Combined production cost for the first 6 widgets: $12 + $20 + $16 + $32 + $22 + $48 = **$150**.

b. **WTP and WTA:**

- **Firm A Max WTP (3rd Permit):** Profit from 3rd widget = $60 - $38 = **$22**.
- **Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = $60 - $48 = **$12**.

c. **After-Trade Total Cost:** Cost is minimized by using the 6 permits on the lowest marginal costs available: 12, 16, 20, 22, 32, and 38. New Total Cost: **$140**.

Version C Solutions

Question 22: Bob & Linda

a. **Absolute Advantage (Burgers): Linda** has the absolute advantage because she can produce **50 burgers** per shift, which is more than Bob's **25 burgers**.

b. **Bob's Opportunity Cost of 1 Burger:** Bob gives up 50 cookies to make 25 burgers. Calculation: $\frac{50\text{ cookies}}{25\text{ burgers}}$ = 2 cookies per burger

c. **Comparative Advantage (Cookies): Linda** has the comparative advantage. Linda's OC of 1 Cookie is $\frac{1}{3}$ burger (50/150), while Bob's OC is $\frac{1}{2}$ burger (25/50).

d. **Range of Prices (Terms of Trade):** Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 3 cookies). Range: **Between 2 and 3 cookies per burger**.

Question 23: Pollution Permits

a. **Initial Total Cost:** Combined production cost for the first 6 widgets: $15 + $30 + $20 + $35 + $25 + $60 = **$185**.

b. **WTP and WTA:**

- **Firm A Max WTP (3rd Permit):** Profit from 3rd widget = $70 - $45 = **$25**.
- **Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = $70 - $60 = **$10**.

c. **After-Trade Total Cost:** Cost is minimized by using the 6 permits on the lowest marginal costs available: 15, 20, 25, 30, 35, and 45. New Total Cost: **$170**.

Version D Solutions

Question 22: Bob & Linda

a. **Absolute Advantage (Burgers): Linda** has the absolute advantage because she can produce **20 burgers** per shift, which is more than Bob's **15 burgers**.

b. **Bob's Opportunity Cost of 1 Burger:** Bob gives up 60 cookies to make 15 burgers. Calculation: $\frac{60 \text{ cookies}}{15 \text{ burgers}}$ = 4 cookies per burger

c. **Comparative Advantage (Cookies): Bob** has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{4}$ burger (15/60), while Linda's OC is $\frac{1}{2}$ burger (20/40).

d. **Range of Prices (Terms of Trade):** Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 4 cookies). Range: **Between 2 and 4 cookies per burger**.

Question 23: Pollution Permits

a. **Initial Total Cost:** Combined production cost for the first 6 widgets: \$20 + \$30 + \$25 + \$35 + \$30 + \$65 = **\$205**.

b. **WTP and WTA:**

- **Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$80 - \$40 = **\$40**.
- **Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$80 - \$65 = **\$15**.

c. **After-Trade Total Cost:** Cost is minimized by using the 6 permits on the lowest marginal costs available: 20, 25, 30, 30, 35, and 40. New Total Cost: **\$180**.

Version E Solutions

Question 22: Bob & Linda

a. **Absolute Advantage (Burgers): Linda** has the absolute advantage because she can produce **60 burgers** per shift, which is more than Bob's **30 burgers**.

b. **Bob's Opportunity Cost of 1 Burger:** Bob gives up 90 cookies to make 30 burgers. Calculation: $\frac{90 \text{ cookies}}{30 \text{ burgers}}$ = 3 cookies per burger

c. **Comparative Advantage (Cookies): Bob** has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{3}$ burger (30/90), while Linda's OC is $\frac{1}{2}$ burger (60/120).

d. **Range of Prices (Terms of Trade):** Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 3 cookies). Range: **Between 2 and 3 cookies per burger**.

Question 23: Pollution Permits

a. **Initial Total Cost:** Combined production cost for the first 6 widgets: \$5 + \$15 + \$10 + \$20 + \$15 + \$40 = **\$105**.

b. **WTP and WTA:**

- **Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$50 - \$25 = **\$25**.
- **Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$50 - \$40 = **\$10**.

c. **After-Trade Total Cost:** Cost is minimized by using the 6 permits on the lowest marginal costs available: 5, 10, 15, 15, 20, and 25. New Total Cost: **\$90**.